Fourier transforms

Consider the Fourier integral representation of the function $f(x)$ given by

$$f(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t)\, e^{is(x-t)}dt\, ds$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{isx}\left[\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)\, e^{-ist)}dt\right]ds$$

Let $F(s)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{-ist}dt$ ------(1)

Then $f(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} F(s)e^{isx}ds$ -----(2)

The integral defined by (1) is called the Fourier transform of the function $f(x)$ and is denoted by $F\{f(x)\}$. Given $F(s)=F\{f(x)\}$, the formula (2) defined $f(x)$, which is called the inverse Fourier transform of $F(s)$ and is denoted by $F^{-1}\{(F(s)\}$.

Note (1): A function $f(x)$ is said to be self-reciprocal under Fourier transforms if $F\{f(x)\}=F(s)$.

Note (2): If $F\{f(x)\}=F(s)$, then $f(x)=F^{-1}\{F(s)\}$ is called spectral representation of $F(s)$ and $F(s)$ is called spectral density of $f(x)$. Here s is called the frequency of the transform. The graph of $|F(s)|$ is called amplitude spectrum of $f(x)$ and $|F(s)|^2$ is called energy of the spectrum.

Properties of the Fourier transforms:

(1) Fourier transform is linear
i.e., if c_1 and c_2 are constants then
$F\{c_1 f(x)+c_2 g(x)\}=c_1F\{f(x)\}+c_2F\{g(x)\}$

Proof: Follows from definition and linearity of the integral.

(2) If $F\{f(x)\} = F(s)$ then $F\{e^{iax} f(x)\} = F(s-a)$

Proof: Consider $F(s-a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i(s-a)t} dt$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist} e^{iat} dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(f(t) e^{iat} \right) e^{-ist} dt$$

$$= F\{e^{iax} f(x)\}$$

(3) If $F\{f(x)\} = F(s)$ then $F\{f(x-a)\} = e^{-isa} F(s)$

Proof: Exercise

(4) If $F\{f(x)\} = F(s)$ then $F\{f(ax)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$

Proof: Exercise

(5) If $F\{f(x)\} = F(s)$ then

(i) $F\{f(x)\cos ax\} = \frac{1}{2}\left(F(s+a) + F(s-a)\right)$

(ii) $F\{f(x)\sin ax\} = \frac{i}{2}\left(F(s+a) - F(s-a)\right)$

(iii) $F\left\{\overline{f(-x)}\right\} = \overline{F(s)}$

(iv) $F\{f(-x)\} = F(-s)$

(v) $F\left\{\overline{f(x)}\right\} = \overline{F(-s)}$

(vi) $F\{x^n f(x)\} = i^n \frac{d^n}{ds^n} F(s)$

(vii) $F\{f^{(n)}(x)\} = (is)^n F\{f(x)\}$

Convolution:

For functions $f(x)\ \&\ g(x)$, we define the convolution of $f(x)\ \&\ g(x)$, denoted by $(f*g)(x)$ as $(f*g)(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)g(x-t)dt$; provided the integral exists.

Note that $f*g=g*f$.

Convolution Theorem:

$F\{(f*g)(x)\}=F\{f(x)\}F\{g(x)\}$.

Proof:

Consider

$$F\{(f*g)(x)\}=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}(f*g)(x)e^{-isx}dx$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(t)g(x-t)dt\right)e^{-isx}dx$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(t)e^{-ist}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}g(x-t)e^{-is(x-t)}dx\right)dt$$

Let $x-t=u$. Then

$$F\{(f*g)(x)\}=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(t)e^{-ist}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}g(u)e^{-is(u)}du\right)dt$$

$$=\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(t)e^{-ist}dt\right)\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}g(u)e^{-is(u)}du\right)$$

$$=F\{f(x)\}F\{g(x)\}.$$

Parseval's Identity:

If $F\{f(x)\}=F(s)$ then $\int_{-\infty}^{\infty}|f(x)|^2\,dx=\int_{-\infty}^{\infty}|F(s)|^2\,ds$.

Proof:

$If\ \ F\{f(x)\} = F(s), F\{g(x)\} = G(s), then$

$F(s)G(s) = F\{f(x) * g(x)\}\ \ or$

$f(x) * g(x) = F^{-1}\{F(s)G(s)\}$

$i.e., \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)g(x-t)dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s)G(s)e^{isx}ds$

For $x = 0$, we get

$$\int_{-\infty}^{\infty} f(t)g(-t)dt = \int_{-\infty}^{\infty} F(s)G(s)ds$$

$Let\ g(x) = \overline{f(-x)}\ or\ g(-x) = \overline{f(x)}.$

$Then G(s) = F\{g(x)\} = F\{\overline{f(-x)}\} = \overline{F\{f(x)\}} = \overline{F(s)}.$

$$\therefore \int_{-\infty}^{\infty} f(t)g(-t)dt = \int_{-\infty}^{\infty} f(t)\overline{f(t)}dt = \int_{-\infty}^{\infty} F(s)\overline{F(s)}ds$$

$$\therefore \int_{-\infty}^{\infty} |f(t)|^2\, dt = \int_{-\infty}^{\infty} |F(s)|^2\, ds$$

Problems:

1) Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$. Hence deduce that

$$\int_{-\infty}^{\infty} \frac{\sin t}{t} dt = \frac{\pi}{2}\ and\ \int_{-\infty}^{\infty} \left(\frac{\sin t}{t}\right)^2 dt = \frac{\pi}{2}.$$

Solution:

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{-isx}dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} 1e^{-isx}dx = \frac{1}{\sqrt{2\pi}} \left(\frac{e^{-isx}}{-is}\right)_{-a}^{a} = \sqrt{\frac{2}{\pi}} \frac{\sin as}{s} = F(s)$$

$$\therefore f(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}F(s)e^{isx}=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\sqrt{\frac{2}{\pi}}\frac{\sin as}{s}(\cos sx+i\sin sx)ds$$

$$=\frac{2}{\pi}\int_{0}^{\infty}\frac{\sin as}{s}\cos sxds \quad \left[\because \frac{\sin as}{s}\sin sx \text{ is an odd function of } s\right]$$

$$\therefore \int_{0}^{\infty}\frac{\sin as}{s}\cos sxds=\frac{\pi}{2}f(x)=\begin{cases}0, & |x|>a\\ \frac{\pi}{2}, & |x|<a\\ \frac{1}{2}\left(\frac{\pi}{2}+0\right)=\frac{\pi}{4}, & |x|=a\end{cases}.$$

For $x=0,\ f(x)=1.$

$$\Rightarrow \int_{0}^{\infty}\frac{\sin as}{s}ds=\frac{\pi}{2}.$$

Let $as=t$ or $s=\frac{t}{a}\therefore ds=\frac{dt}{a}.$

On substitution, we get

$$\int_{0}^{\infty}\frac{\sin t}{t/a}\frac{dt}{a}=\frac{\pi}{2}$$

or $$\int_{0}^{\infty}\frac{\sin t}{t}dt=\frac{\pi}{2}.$$

By Parseval's identity,

$$\int_{-\infty}^{\infty}|f(x)|^2\,dx=\int_{-\infty}^{\infty}|F(s)|^2\,ds$$

i.e., $$\int_{-a}^{a}1dx=\int_{-\infty}^{\infty}\frac{2}{\pi}\left(\frac{\sin as}{s}\right)^2ds=\frac{4}{\pi}\int_{0}^{\infty}\left(\frac{\sin as}{s}\right)^2ds$$

$$\int_{0}^{\infty}\left(\frac{\sin as}{s}\right)^2ds=\frac{\pi a}{2}$$

For $a=1,$ we get $\int_{0}^{\infty}\left(\frac{\sin t}{t}\right)^2ds=\frac{\pi}{2}.$

2) Find the Fourier transform of $e^{-a|x|}, a>0$ and hence evaluate $\int_0^\infty \frac{\cos xt}{a^2+t^2}dt$ and $F\{xe^{-a|x|}\}$

$$F\{e^{-a|x|}\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-a|x|}e^{-isx}dx = \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-ax}\cos sxdx$$

$$= \sqrt{\frac{2}{\pi}}\frac{e^{-ax}}{a^2+s^2}\left(-a\cos sx + s\sin sx\right)_0^\infty$$

$$= \sqrt{\frac{2}{\pi}}\ (0 - \frac{e^{-o}}{a^2+s^2}(-a)) = \sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2} = F(s)$$

$$\therefore f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} F(s)e^{isx}ds$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \frac{a}{a^2+s^2}(\cos sx + i\sin sx)ds$$

$$= \frac{2}{\pi}\int_0^\infty \frac{a\cos sx}{a^2+s^2}ds$$

$$\therefore \int_0^\infty \frac{\cos sx}{a^2+s^2}ds = \int_0^\infty \frac{\cos xt}{a^2+t^2}dt = \frac{\pi\ f(x)}{2a} = \frac{\pi}{2a}e^{-a|x|}$$

Since

$$F\{x^n f(x)\} = (-i)^n \frac{d^n}{ds^n}F(s)$$

$$for\ n=1, F\{xe^{-a|x|}\} = (-i)^1\frac{d}{ds}F\{e^{-a|x|}\} = -i\sqrt{\frac{2}{\pi}}\frac{d}{ds}\frac{a}{a^2+s^2} = i\sqrt{\frac{2}{\pi}}\left(\frac{2as}{(a^2+s^2)^2}\right)$$

3) Find the Fourier transform of $e^{-a^2x^2}, a>0$ and hence show that $e^{-\frac{x^2}{2}}$ is self-reciprocal under Fourier transform.

$$F\{e^{-a^2x^2}\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-a^2x^2}e^{-isx}\,dx = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(a^2x^2+isx)}\,dx$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\left(ax+\frac{is}{2a}\right)^2-\frac{s^2}{4a^2}}\,dx$$

$$= \frac{e^{-\frac{s^2}{4a^2}}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\left(ax+\frac{is}{2a}\right)^2}\,dx$$

Let $ax+\frac{is}{2a}=t$ then $dx=\frac{dt}{a}$

$$= \frac{e^{-\frac{s^2}{4a^2}}}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-t^2}\frac{dt}{a} = \frac{2e^{-\frac{s^2}{4a^2}}}{a\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-t^2}\,dt$$

$t^2 = y$ or $t=\sqrt{y}$

Put

$\therefore dt = \frac{1}{2}y^{\frac{-1}{2}}dy$

$$= 2\frac{e^{-\frac{s^2}{4a^2}}}{\sqrt{2\pi}}\int_0^{\infty}\frac{e^{-y}}{2}y^{-\frac{1}{2}}\,dy = \frac{e^{-\frac{s^2}{4a^2}}}{a\sqrt{2\pi}}\Gamma\left(\frac{1}{2}\right) = \frac{e^{-\frac{s^2}{4a^2}}}{a\sqrt{2\pi}}\sqrt{\pi} = \frac{e^{-\frac{s^2}{4a^2}}}{a\sqrt{2}}$$

For $a=\frac{1}{\sqrt{2}}$ or $\sqrt{2}a=1$

$$F\{e^{-a^2x^2}\} = F\{e^{-x^2/2}\} = e^{-s^2/2}$$

$\therefore e^{-s^2/2}$ is self-reciprocal under Fourier transform.

4) Find the Fourier transform of $f(x)=\begin{cases}1-|x| & |x|\le 1\\ 0 & |x|>1\end{cases}$ and hence evaluate $\int_0^{\infty}\left(\frac{\sin x}{x}\right)^2 dx$

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)e^{-isx}dx = \frac{1}{\sqrt{2\pi}}\int_{-1}^{1}(1-|x|)(\cos sx - i\sin sx)\,dx$$

$$= \sqrt{\frac{2}{\pi}}\int_{0}^{1}(1-x)\cos sx\,dx \quad \because (1-|x|)\sin sx \text{ is an odd function of } x.$$

$$= \sqrt{\frac{2}{\pi}}\left((1-x)\frac{\sin sx}{s} - \frac{\cos sx}{s^2}\right)_0^1 = \sqrt{\frac{2}{\pi}}\frac{(1-\cos s)}{s^2} = F(s)$$

$$\therefore f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} F(s)e^{isx}ds$$

$$= \frac{1}{\sqrt{2\pi}}\sqrt{\frac{2}{\pi}}\int_{-\infty}^{\infty}\frac{(1-\cos s)}{s^2}(\cos sx + i\sin sx)ds$$

$$= \frac{2}{\pi}\int_{0}^{\infty}\frac{(1-\cos s)}{s^2}\cos sx\,ds = \frac{\pi}{2}f(x) = \begin{cases}\frac{\pi}{2}(1-|x|) & |x|\le 1\\ 0 & |x|>1\end{cases}$$

$$For\ x=0,\ \int_0^\infty \frac{(1-\cos sx)}{s^2}ds = \frac{\pi}{2}(1-0) = \frac{\pi}{2}$$

$$\Rightarrow \int_0^\infty \frac{2\sin^2(s/2)}{s^2}ds = \frac{\pi}{2}$$

$$Let\ s/2 = t\ or\ s = 2t \quad ds = 2dt$$

$$\int_0^\infty \frac{2\sin^2 t}{4t^2}\,2dt = \frac{\pi}{2}\ or\ \int_0^\infty\left(\frac{\sin t}{t}\right)^2 dt = \frac{\pi}{2}$$

Exercises:

1. Obtain the Fourier transform of
$$f(x) = \begin{cases}1-x^2 & |x|\le 1\\ 0 & |x|>1\end{cases}$$ and hence evaluate $\int_0^\infty\left(\frac{x\cos x - \sin x}{x^3}\right)\cos(x/2)dx$
2. Find the Fourier transform of $f(x) = \begin{cases}|x| & |x|\le a\\ 0 & |x|>a>0\end{cases}$
3. Find the Fourier transform of $f(x) = \begin{cases}a^2-x^2 & |x|\le a\\ 0 & |x|>a\end{cases}$

Fourier Cosine and Sine transforms

Consider the Fourier cosine integral representation of a function f(x)

$$f(x)=\frac{2}{\pi}\int_0^\infty\int_0^\infty f(t)\cos st\cos sx\,dtds\,,\quad x\geq 0$$

$$=\sqrt{\frac{2}{\pi}}\int_0^\infty\left[\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos st\;dt\right]\cos sx\,ds$$

$$Let\;\; F_c\{f(x)\}=\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos st\,dt\;=F_c(s)....(1)$$

$$Then\;\; f(x)=\sqrt{\frac{2}{\pi}}\int_0^\infty F_c\{f(x)\}\cos sx\,ds\qquad .....(2)$$

The transform $F_c\{f(x)\}$ defined by (1) is called the Fourier cosine transform of f(x).The formula (2) is called the inverse Fourier cosine transform of $F_c\{f(x)\}=F_c(s)$ and is denoted by $f(x)=F_c^{-1}\{F_c(s)\}$. Similarly, using the Fourier sine integral representation of f(x) given by $f(x)=\frac{2}{\pi}\int_0^\infty\int_0^\infty f(t)\sin st\sin sx\,dt\,ds,$ we can define the Fourier sine transform of f(x) denoted by $F_s\{f(x)\}$ as $F_s\{f(x)\}=\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\sin st\,dt=F_s(s)$

Then the inverse Fourier sine transform of $F_s(s)$ is defined as

$$f(x)=F_s^{-1}\{F_s(s)\}=\sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\sin sx\,ds$$

Definition: A function f(x) is said to be self reciprocal under Fourier cosine (sine) transform if $F_c\{f(x)\}=f(s)\qquad\left(Fs\{f(x)\}=f(s)\right)$

Properties of Fourier cosine/ sine transforms

(1) Both Fourier cosine and sine transforms are linear.

$$F_c\{c_1 f(x) + c_2 g(x)\} = c_1 F_c\{f(x)\} + c_2 F_c\{g(x)\}$$

$$F_s\{c_1 f(x) + c_2 g(x)\} = c_1 F_s\{f(x)\} + c_2 F_s\{g(x)\} \text{ where } c_1 \text{ and } c_2 \text{ are constants}$$

(2). $$F_c\{f(x)\cos ax\} = \frac{1}{2}\left(F_c(s+a) + F_c(s-a)\right)$$

$$F_c\{f(x)\sin ax\} = \frac{1}{2}\left(F_s(s+a) + F_s(a-s)\right)$$

(3). $$F_s\{f(x)\cos ax\} = \frac{1}{2}\left(F_s(s+a) + F_s(s-a)\right)$$

$$F_s\{f(x)\sin ax\} = \frac{1}{2}\left(F_c(s-a) - F_s(s+a)\right)$$

The proof of (2) and (3) follows directly from trigonometric identities.

(4). $$F_c\{f(ax)\} = \frac{1}{a}F_c(s/a)$$

$$F_s\{f(ax)\} = \frac{1}{a}F_s(s/a)$$

(5). If $f(x) \to 0$ as $x \to \infty$, then

$$F_c\{f'(x)\} = -\sqrt{\frac{2}{\pi}}f(0) + s\,F_s(s)$$

$$F_s\{f'(x)\} = -s\,F_c(s)$$

Proof : Consider $F_c\{f'(x)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty f'(x)\cos sx\,dx$

Integrating by parts, we get

$$F_C\{f'(x)\} = \sqrt{\frac{2}{\pi}}\left(f(x)\cos sx\,|_0^\infty - \int_0^\infty f(x)\sin sx\,(-s)dx\right)$$

$$= \sqrt{\frac{2}{\pi}}\left(0 - f(0) + s\int_0^\infty f(x)\sin sx\,dx\right)$$

$$= sF_S\{f(x)\} - \sqrt{\frac{2}{\pi}}f(0) \text{ provided } f(x) \to 0 \text{ as } x \to \infty$$

Also,

$$F_S\{f'(x)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty f'(x)\sin sx\ dx$$

Integrating by parts, we get

$$F_S\{f'(x)\} = \sqrt{\frac{2}{\pi}}\left(f(x)\sin sx\,|_0^\infty - \int_0^\infty f(x)\cos sx\ (s)dx \right)$$

$$= \sqrt{\frac{2}{\pi}}\left(0 - 0 - \sqrt{\frac{\pi}{2}}sF_C(s) \right) = -sF_C(s).$$

(6) $F_C\{f''(x)\} = -\sqrt{\frac{2}{\pi}}f'(0) - s^2F_C(s)$

$$F_S\{f''(x)\} = \sqrt{\frac{2}{\pi}}sf(0) - s^2F_S(s)$$

provided $f(x)$ and $f'(x) \to 0$ as $x \to \infty$

(7) $F_C\{xf(x)\} = \frac{dF_S}{ds}$ and $F_S\{xf(x)\} = -\frac{dF_C}{ds}$

(8) *If* $F_C\{f(x)\} = F_C(s), F_C\{g(x)\} = G_C(s),$
$F_S\{f(x)\} = F_S(s)$ and $F_S\{g(x)\} = G_S(s)$ exist,

then $\int_0^\infty F_C(s)\ G_C(s)\ ds = \int_0^\infty F_S(s)\ G_S(s)\ ds = \int_0^\infty f(x)g(x)dx$

and $\int_0^\infty |F_C(s)|^2\ ds = \int_0^\infty |F_C(s)|^2 ds = \int_0^\infty |f(x)|^2 dx$

which is called Parsival's identity.

Problems:

(1) Find $\mathrm{F_C}\left\{e^{-ax}\right\}$, $\mathrm{F_S}\left\{e^{-ax}\right\}$ and hence find $\mathrm{F_C}\left\{xe^{-ax}\right\}$ and $\mathrm{F_S}\left\{xe^{-ax}\right\}$

Solution: By definition,

$$\mathrm{F_C}\left\{e^{-ax}\right\} = \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-ax}\cos sx\ dx$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{e^{-ax}}{a^2+s^2}\left(-a\cos sx + s\sin sx\right)\right]_0^\infty = \sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}, a \geq 0$$

$$\mathrm{F_S}\left\{e^{-ax}\right\} = \sqrt{\frac{2}{\pi}}\int_0^\infty e^{-ax}\sin sx\ dx$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{e^{-ax}}{a^2+s^2}\left(-a\sin sx - s\cos sx\right)\right]_0^\infty = \sqrt{\frac{2}{\pi}}\frac{s}{a^2+s^2},\ a \geq 0.$$

Therefore,

$$\mathrm{F_C}\left\{xe^{-ax}\right\} = \frac{d}{ds}F_S\left\{e^{-ax}\right\} = \frac{d}{ds}\left(\sqrt{\frac{2}{\pi}}\frac{s}{a^2+s^2}\right) = \sqrt{\frac{2}{\pi}}\frac{a^2-s^2}{(a^2+s^2)^2}$$

$$\mathrm{F_S}\left\{xe^{-ax}\right\} = -\frac{d}{ds}F_C\left\{e^{-ax}\right\} = -\frac{d}{ds}\left(\sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}\right) = \sqrt{\frac{2}{\pi}}\frac{2as}{(a^2+s^2)^2}$$

2). Find $F_c\left\{e^{-a^2x^2}\right\}$ & $F_s\left\{xe^{-a^2x^2}\right\}$.

Solution : We have from definition,

$$F_c\left\{e^{-a^2x^2}\right\}=\sqrt{\frac{2}{\pi}}\int_0^\infty e^{-a^2x^2}\cos sx\,dx=\sqrt{\frac{2}{\pi}}\int_0^\infty e^{-a^2x^2}e^{-isx}dx=F\left\{e^{-a^2x^2}\right\}$$

$$=\frac{1}{\sqrt{2}a}e^{-\frac{s^2}{4a^2}}$$

$$\therefore F_s\left\{xe^{-a^2x^2}\right\}=-\frac{d}{ds}F_c\left\{e^{-a^2x^2}\right\}=-\frac{d}{ds}\left(\frac{1}{\sqrt{2}a}e^{-\frac{s^2}{4a^2}}\right)=\frac{s}{2\sqrt{2}a^3}e^{-\frac{s^2}{4a^2}}$$

Note that

for $a=\frac{1}{\sqrt{2}}, F_c\left\{e^{-a^2x^2}\right\}=F_c\left\{e^{-\frac{x^2}{2}}\right\}=e^{-\frac{s^2}{2}}$

& $F_s\left\{xe^{-a^2x^2}\right\}=F_s\left\{xe^{-\frac{x^2}{2}}\right\}=xe^{-\frac{s^2}{2}}.$

$\therefore e^{-\frac{x^2}{2}}$ is self reciprocal under Fourier cosine transform and

$xe^{-\frac{x^2}{2}}$ is self reciprocal under Fourier sine transform .

3) Find $F_c\left\{x^{a-1}\right\}$ & $F_s\left\{x^{a-1}\right\}, 0<a<1.$

Consider $\int_0^\infty x^{a-1}e^{-isx}dx;$

Let $isx=y$ or $dx=\frac{dy}{is}$. Then

$$\int_0^\infty x^{a-1}e^{-isx}dx=\frac{1}{(is)^a}\int_0^\infty y^{a-1}e^{-y}dy=\frac{\Gamma(a)}{i^as^a}=(-i)^a\frac{\Gamma(a)}{s^a}=\left(e^{-i\frac{\pi}{2}}\right)^a\frac{\Gamma(a)}{s^a}$$

$$=\cos\left(\frac{a\pi}{2}\right)\frac{\Gamma(a)}{s^a}-i\sin\left(\frac{a\pi}{2}\right)\frac{\Gamma(a)}{s^a}$$

Thus

$$\int_0^\infty x^{a-1}\cos sxdx - i\int_0^\infty x^{a-1}\cos sxdx = \cos\left(\frac{a\pi}{2}\right)\frac{\Gamma(a)}{s^a} - i\sin\left(\frac{a\pi}{2}\right)\frac{\Gamma(a)}{s^a}$$

Equating real and imaginary parts, we get

$$F_c\left\{x^{a-1}\right\} = \sqrt{\frac{2}{\pi}}\int_0^\infty x^{a-1}\cos sxdx = \sqrt{\frac{2}{\pi}}\frac{\Gamma(a)}{s^a}\cos\left(\frac{a\pi}{2}\right) \text{ and}$$

$$F_s\left\{x^{a-1}\right\} = \sqrt{\frac{2}{\pi}}\int_0^\infty x^{a-1}\sin sxdx = \sqrt{\frac{2}{\pi}}\frac{\Gamma(a)}{s^a}\sin\left(\frac{a\pi}{2}\right).$$

Note:

$$\text{For } a = \frac{1}{2}, F_c\left\{x^{a-1}\right\} = F_c\left\{\frac{1}{\sqrt{x}}\right\} = \frac{1}{\sqrt{s}} = F_s\left\{\frac{1}{\sqrt{x}}\right\} \quad \left(\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}\right)$$

$\therefore \frac{1}{\sqrt{x}}$ is self reciprocal under Fourier sine and cosine transforms.

Note:

Consider $F_c\{f(x)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos sxdx = F_c(s).$

Then $f(x) = F_c^{-1}\{F_c(s)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty F_c(s)\cos sxds$ ------(1)

Interchanging s and x in (1), we get

$$f(s) = \sqrt{\frac{2}{\pi}}\int_0^\infty F_c(x)\cos sxdx = F_c\{F_c(s)\}$$

Similarly, if $F_s\{f(x)\} = F_s(s)$ then, $F_s\{F_s(s)\} = f(s).$

4) Find

$F_c\left\{\frac{1}{1+x^2}\right\}$ and $F_s\left\{\frac{x}{1+x^2}\right\}$.

Solution: We have shown that

$$F_c\left\{e^{-ax}\right\}=\sqrt{\frac{2}{\pi}}\frac{a}{a^2+s^2}=F_c(s).$$

$$\therefore F_c\left\{F_c(x)\right\}=F_c\left\{\sqrt{\frac{2}{\pi}}\frac{a}{a^2+x^2}\right\}=e^{-as}.$$

Or $F_c\left\{\frac{a}{a^2+x^2}\right\}=\sqrt{\frac{\pi}{2}}e^{-as}$.

For $a=1, F_c\left\{\frac{1}{1+s^2}\right\}=\sqrt{\frac{\pi}{2}}e^{-s}$.

$$F_s\left\{\frac{x}{1+x^2}\right\}=-\frac{d}{ds}\sqrt{\frac{\pi}{2}}e^{-s}=\sqrt{\frac{\pi}{2}}e^{-s}.$$

Exercises:

1) Find the Fourier cosine transform of $e^{-ax}\cos ax, e^{-ax}\sin ax$ and hence find the Fourier cosine transforms of $\frac{1}{\left(x^4+k^4\right)}$ and $\frac{x^2}{\left(x^4+k^4\right)}$.
2) Find the Fourier sine transform of $\frac{1}{x}$ and $\frac{e^{-ax}}{x}, a>0$.